Complex Systems Intuitively

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How do we truly learn? By deconstructing. By doing something unexpected or something that demands pure creativity.

I recently read a Springer textbook. I came across randomly but I can assure it changed the way I see the world : ‘An Introduction to Complex Systems : Making Sense of a Changing World’ by Joe Tranquillo, Professor at Bucknell University. Just by reading the 30-pages introduction, I knew I found a book that would amaze me.

It inspired me to create some random things, to explore.

I built a random-walk web crawler, because why not. It’s not usefull and that’s the whole purpose. It’s nearly art. The more it’s useless, the more it’s art. It’s a bot that goes from website to website and jumps on a random link on the current website with some constraints. We can see patterns emerging.

Constraints : we can’t go on a domaine that has been visited X times, not funny otherwise, a bit of change is better. the bot is allowed to backtrack if it reaches a dead end. And we avoid some pages such as login, privacy, support, contact etc.

This is a simple exemple of how random structures can emerge with nearly nothing. I love this. I think theses could be showed to some kids to have them develop intuition about internet, randomness and complex systems.

When reading the book of Tranquillo, I understood that : everything can be mapped with the framework of complex systems.

I’ve since started other “weird” projects. I’m proud of their uselessness. they are intentionally purposeless, which, paradoxically, is what makes them valuable to me.

Take my project weathee. It generates visuals based on live meteorological data and random matrix theory. It’s a bit simple, but there’s a certain magic in it when you don’t know the underlying mechanics.

By building these, I want to sharpen my intuition and see how interdisciplinary fields collide: computer networks, evolutionnary algorithms, stochastic process, art, and physics. I originally wanted to use a physical thermometer with a Raspberry Pi to bridge the digital and the physical, but I did not had time, so I settled for a weather API.